

Self-Powered LLM Modality Expansion for Large Speech-Text Models

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Abstract

Large language models (LLMs) exhibit remarkable performance across diverse tasks, indicating their potential for expansion into large speech-text models (LSMs) by integrating speech capabilities. Although unified speech-text pre-training and multimodal data instruction-tuning offer considerable benefits, these methods generally entail significant resource demands and tend to overfit specific tasks. This study aims to refine the use of speech datasets for LSM training by addressing the limitations of vanilla instruction tuning. We explore the instruction-following dynamics within LSMs, identifying a critical issue termed *speech anchor bias*—a tendency for LSMs to over-rely on speech inputs, mistakenly interpreting the entire speech modality as directives, thereby neglecting textual instructions. To counteract this bias, we introduce a self-powered LSM that leverages augmented automatic speech recognition data generated by the model itself for more effective instruction tuning. Our experiments across a range of speech-based tasks demonstrate that self-powered LSM mitigates speech anchor bias and improves the fusion of speech and text modalities in LSMs. Data, code and scripts are freely available at <https://anonymous.4open.science/r/Self-powered-LSM-1EE3>.

1 Introduction

Recent advances in large language models (LLMs) (Brown et al., 2020; Touvron et al., 2023a; Anil et al., 2023) have achieved remarkable performance across numerous tasks (OpenAI, 2023). To further extend the capabilities of LLMs, many efforts (Liu et al., 2023a; Zhang et al., 2023b; Rubenstein et al., 2023) have been made to integrate LLMs with multi-modal encoders, aiming to endow models with multimodal capabilities. Speech, as a typical modality, shares similar semantics with text and has become a widely studied area.

To expand the speech capabilities of LLMs, several studies adopt multitask learning (Chu et al., 2023) or unified speech-text pre-training (Wang et al., 2023c; Rubenstein et al., 2023). While yielding notable performance, these methods require extensive computational resources. To address this, Tang et al. (2023) construct multimodal speech instruction datasets to fine-tune LLMs for developing large speech-text models (LSMs). However, vanilla datasets often reveal LSMs’ deficiencies in following instructions (Wang et al., 2023a; Pan et al., 2023). To avoid task-specific overfitting and enhance generalizability, careful development of datasets and methodologies is necessary.

Given the prevalence of automatic speech recognition (ASR) datasets, using this data to build multimodal speech instruction datasets is straightforward. Thus, we question whether LSMs can be fully developed solely by collecting this type of training data. To achieve this, this study first explores the attention dynamics between speech inputs and instructions, identifying inherent training challenges presented by vanilla ASR training. Our analysis reveals a pronounced tendency for LSMs trained with speech instructional data to overly concentrate on speech, neglecting instructions. This predisposition causes LSMs to mistakenly interpret the entire speech modality as instruction, limiting the model’s responses to scenarios that closely mirror its training data. We define this phenomenon as *speech anchor bias*.

Motivated by this finding, we argue that emphasizing the role of instructions during the training process is crucial for enhancing the instruction-following ability of LSMs. To achieve this, we introduce a novel self-powered LSM that utilizes augmented instructional data generated by the model itself to facilitate the expansion of modality capabilities. Experiments conducted on ASR, speech translation (ST), speech-language understanding (SLU), and question answering (QA) demonstrate

that our method, even when applied solely to ASR datasets, effectively improves LSM performance across diverse tasks, thereby alleviating speech anchor bias and enhancing the integration of speech and text modalities in LSMs

The **main contributions** of this paper are:

- We unveil a critical issue in LSM training—*speech anchor bias*, where direct instruction tuning causes excessive reliance on speech inputs, diminishing the model’s competence with instructions.
- We introduce an innovative self-powered LSM, which leverages self-generated data to rapidly enhance the speech modality capabilities of LLMs.
- Further analysis confirms that our method effectively reduces speech anchor bias, achieving better alignment between speech and text.
- We publicly release our self-powered augmentation dataset, hoping it will benefit further research within the community.

2 Related Work

2.1 Pretraining LSMs for Modality Extension

Recent advancements in pretraining LSMs can be divided into two categories. The first focuses on self-supervised speech representation learning without text supervision, enabling unsupervised learning from extensive unlabeled speech data (Mohamed et al., 2022; Wu et al., 2023a). This approach also facilitates the generation of high-quality speech tokens (Lakhotia et al., 2021; Popuri et al., 2022; Zhang et al., 2023c). The second approach unifies speech and text within a single model, as exemplified by models like ViOLA (Wang et al., 2023c), AudioPalm (Rubenstein et al., 2023), SpeechGPT (Zhang et al., 2023a), LauraGPT (Chen et al., 2023b), and Qwen-Audio (Chu et al., 2023). These models leverage a shared vocabulary for speech and text, or use a multi-task training framework to continue training the LLM with a speech dataset, thereby enhancing their capabilities across various speech tasks. However, pretraining an LSM requires more resources and complex procedures, making it difficult to quickly adapt to different LLM backbones and rendering it a challenging endeavor.

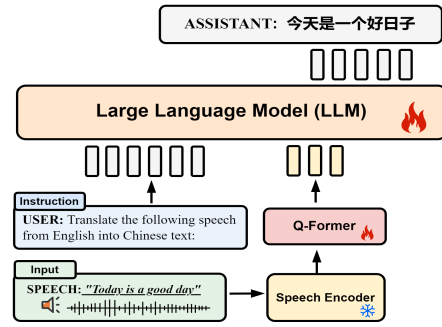


Figure 1: Model architecture of LSM.

2.2 Expand LLMs with Speech Capabilities

Following the advancement of ChatGPT, AudioGPT (Huang et al., 2024) has enabled LLMs to process speech by interacting within task-specific model pipelines. Efforts towards end-to-end integration include aligning speech and text embeddings through connection modules between speech encoders and LLMs (Chen et al., 2023a; Wu et al., 2023b; Yu et al., 2023; Wang et al., 2023a) or integrating LoRA (Hu et al., 2022) into LLMs, thereby enhancing their speech capabilities. Notably, studies by Tang et al. (2023) and Wang et al. (2023a) indicate that directly training LLMs with both speech and target transcripts can lead to overfitting on specific speech tasks, potentially causing modality imbalance in LSM. Despite efforts to address this issue, previous work still requires task-specific training adjustments, limiting efficient adaptation across various LLMs. Our work comprehends the model’s behavior through layer-wise attention and introduces a straightforward training framework to efficiently expand LLMs with speech capabilities.

3 Methodology

3.1 Preliminary

Model Architecture The model architecture of LSM is shown in Figure 1. We use the encoder component of Whisper-small (Radford et al., 2023) as the speech encoder and employ Vicuna-7B-1.5 (Chiang et al., 2023) as the large language model. Q-Former (Li et al., 2023a), serving as the connection module, employs N trainable queries Q within its stacked blocks. The output sequence, integrated with the text instruction, is then fed into the LLM to generate the text response.

Training Objective We consider a set of model parameters parameterized by θ for training. Let s represent the speech input and t its correspond-

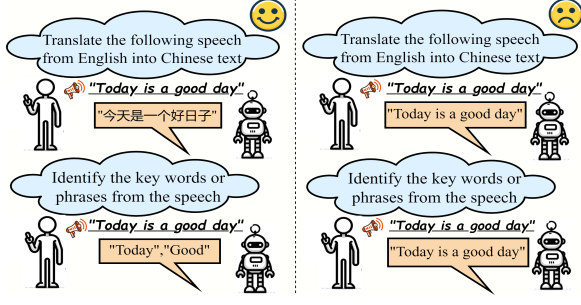


Figure 2: Directly fine-tuning of an LSM with speech instructional data does not enable the acquisition of speech modality expansion capability.

ing target text sentence. The whisper encoder receives the original sequence s as input and produces its contextual representation $\mathbf{H}$. To equip the Q-Former with the ability to handle variable-length general speech inputs, we use the window-level segment strategy (Tang et al., 2023) to segment $\mathbf{H}$ into L -sized window representations and utilize Q-Former at the window level to output textual tokens $\mathbf{Z}$. The log-likelihood training objective for the parallel speech-text pair (s, t) is:

$$\begin{aligned}\hat{\theta} &= \arg \min_{\theta} (-\log P(t|s, i; \theta)) \\ &= \arg \min_{\theta} \left(-\sum_{m=1}^M \log P(t_m | t_{<m}, s, i; \theta) \right),\end{aligned}\quad (1)$$

where M is the length of the target text, t_m is the m -th target token, and i is the text embedding of instruction. The loss is a standard causal language modeling loss, which predicts the next token based on prior tokens. We use the same prompt template as vicuna (Chiang et al., 2023) to keep training consistent and do not compute the loss for the instruction during training (Wang et al., 2023a).

3.2 Modality Imbalance for LSMs

To enhance the speech capabilities of LLMs, a common approach involves fine-tuning these models using multimodal instructional samples and organizing data into a structured format that includes textual instructions, speech inputs, and text responses. However, existing research (Tang et al., 2023; Wang et al., 2023a) has highlighted a significant deficiency in LSM’s ability to adhere to instructions when trained with vanilla speech instructional data. We illustrate this challenge in Figure 2. To investigate the reason behind this modality imbalance during LSM training, we conduct analysis by examining the attention mechanisms across both speech and instructional inputs, exploring information interaction.

Attention Comparison Attention weights is a common interpretation tool that aims to discover the inherent patterns in the attention interaction to ascertain which input vectors contribute (Clark et al., 2019b; Liu et al., 2021). For multi-modal inputs, the features themselves also affect critical token interactions. Meanwhile, numerous studies argue that analyzing the attention mechanism with only attention weights overlooks the effect of the transformed vector (Wiegrefe and Pinter, 2019; Bibal et al., 2022). To gain a more in-depth analysis of attention interaction, we refer the *norm of the weighted transformed vector* (Kobayashi et al., 2020) and define the metric A_j ¹ denoting the average attention score from the j -th input token to the model’s output².

Inspired by Wang et al. (2023b), we introduce quantitative metrics derived from $\mathbf{A}$ to elucidate the information flow from speech inputs and text instructions to the generated sequence. These metrics are expressed as:

$$S_{\text{Instruction}} = \frac{1}{|\mathbf{I}|} \sum_{\mathbf{I}_j \in \mathbf{I}} \mathbf{A}_j, \quad S_{\text{Speech}} = \frac{1}{|\mathbf{Z}|} \sum_{\mathbf{Z}_j \in \mathbf{Z}} \mathbf{A}_j, \quad (2)$$

where $S_{\text{Instruction}}$ and S_{Speech} mean the information flow from instruction part $\mathbf{I}$ and speech part $\mathbf{Z}$ to output sequence. To assess the relative focus on instruction versus speech components within input features, we calculate the proportion η for each layer as $\frac{S_{\text{Instruction}}}{S_{\text{Instruction}} + S_{\text{Speech}}}$. The η value indicates the relative contributions of speech and instruction to model predictions, offering insights into the model’s attention allocation during generation.

Differences between LLMs and LSMs By using quantitative metrics, we conducted a comparative analysis of layer-wise behavior between different types of instruction-tuning in LLMs and Language and LSMs regarding how they handle instruction and source inputs. Detailed training and evaluation information is available in Appendix A.1. Figure 3 shows that after training, LLMs such as Alpaca and Vicuna exhibit dynamic attention shifts: the early layers distribute attention evenly between instructions and source inputs; the middle layers prioritize source information; and the deeper layers

¹The more details of the definition are provided in Appendix A.1.

²During the generation of the m -th token, decoder-only model appends the preceding $m - 1$ tokens to the end of the input sequence as part of its next token prediction process. To simplify our analysis, we focus on computing the scores exclusively for the tokens within the initial input’s length when generating the m -th token.

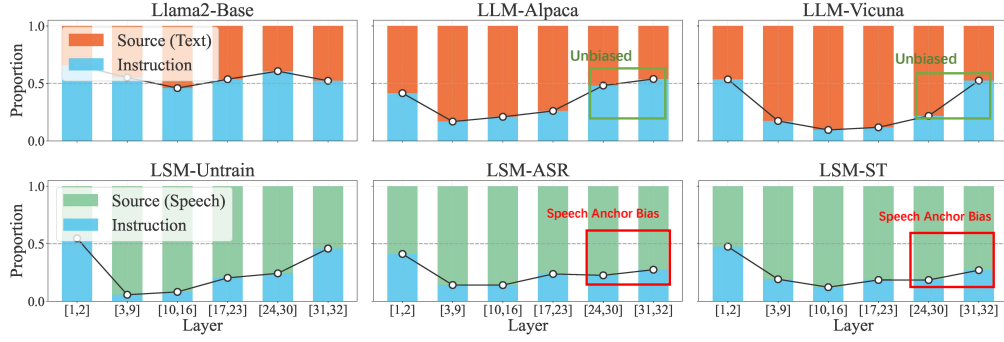


Figure 3: The comparison of the layer-wise behavior in instruction-following LLM versus instruction-ignoring LSM. “Source” refers to text input for LLMs, whereas denotes speech input for LSM. As the layer deepens, the proportion of instructions diminishes in LSM while increasing in LLM. The red borders show that LSMs excessively focus on speech representations and ignore instructions.

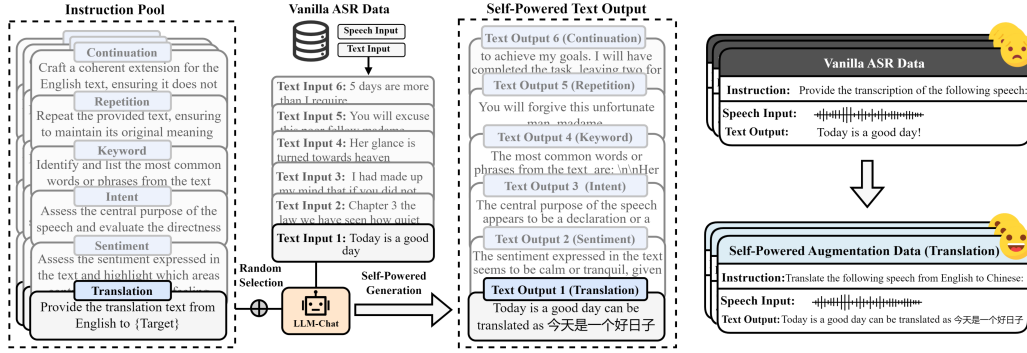


Figure 4: Process of self-powered data augmentation: Self-Powered data is generated by prompting the LLM with instructions alongside the text from the vanilla ASR dataset. The self-powered data is then used to train the LSM.

refocus on instructions, integrating them into the outputs. In contrast, LSMs, after vanilla training, especially those trained on ASR and ST tasks like LSM-ASR and LSM-ST, *consistently favor speech inputs over instructions across all layers*.

Speech Anchor Bias Although LSMs are inherently designed to process dual-modal inputs—speech and text—our understanding reveals a significant bias toward speech inputs during training, resulting in the model’s complete oversight of textual instructions. During inference, this tendency causes LSMs to interpret the entire speech modality as the instructions that appeared in the training stage, limiting the model’s responses to more diverse text instructions. We term this issue “speech anchor bias”, which is the main reason for causing modality imbalance when using LLMs to expand speech capacity for LSMs.

3.3 Modality Expansion via Self-Powered Augmentation

Building on insights from § 3.2, it is imperative to avert modality imbalance during LSM training by

emphasizing the role of instructions in the training process. To achieve this, we propose self-powered LSM, which capitalizes on self-powered data to enhance the speech modality capabilities of LLMs for LSMs. A comprehensive process of our proposed method is illustrated in Figure 4.

Self-Powered Data Generation Given an LSM θ , we delineate it into three components: the speech encoder θ_s , the Q-former θ_q , and LLM θ_l . We organize an instruction pool $\mathcal{I}$ into K distinct tasks $\mathcal{I}_1, \mathcal{I}_2, \dots, \mathcal{I}_K$. Each task is associated with m specific types of instances. For each speech-text paired ASR data (s, t) , we systematically select one task from the instruction pool and randomly employ a specific instruction i_{div} from m instances. This instruction is used along with the text t to prompt θ_l , resulting in the generation of a self-powered text output $\hat{t}$, as described by the following equation:

$$\hat{t} = \arg \max_t P(\hat{t}|t, i_{div}; \theta_l). \quad (3)$$

Subsequently, the vanilla ASR training sample (s, t) is supplanted with the newly generated self-powered augmentation sample $(s, \hat{t}, i_{div})$. It is im-

portant to highlight that any form of instruction instance can contribute to the generation of self-powered text. To diversify and specify the types of instructions, we craft distinctive categories within the speech domain and generate multiple instances of instructions for each category.

Self-Powered Model Training We use constructed self-powered augmentation data for LSM training. During training, the speech encoder is frozen to ensure the stability of speech features while the Q-former and LLM are fine-tuned for alignment. The training objective can be formulated as:

$$\hat{\theta}_q, \hat{\theta}_l = \arg \min_{(\theta_q, \theta_l)} \left(-\log P(\hat{\mathbf{t}}|\mathbf{s}, \mathbf{i}_{div}; \theta_s^*, \theta_q^*, \theta_l^*) \right). \quad (4)$$

By employing self-powered augmentation data, we can significantly enhance the model’s focus on instructions during the training process. In the subsequent section, we provide a theoretical exposition of how self-powered augmentation data effectively avoids modality imbalance during LSM training.

3.4 Discussion

When training LSMs with vanilla speech instructional data, speech anchor bias causes models to disregard textual instructions, potentially skewing the distribution $P(\mathbf{t}|\mathbf{s}, \mathbf{i}; \theta)$ to closely align with $P(\mathbf{t}|\mathbf{s}; \theta)$. Then the training objective is degraded into $-\log P(\mathbf{t}|\mathbf{s}; \theta)$, restricting the model’s response to the tasks within the training dataset.

Conversely, training LSMs with self-powered augmentation data effectively redirects the training objectives to maximize the likelihood of an instruction-constrained distribution and then strategically shifts the model’s focus toward closer alignment with the instruction $\mathbf{i}$.

$$\begin{aligned} \mathcal{L}(\theta) &= -\log P(\mathbf{t}|\mathbf{s}, \mathbf{i}; \theta) \\ &\doteq -\log P(\mathbf{t}|\mathbf{s}; \theta) \quad (\text{Speech Anchor Bias}) \\ &\doteq -\log P(\hat{\mathbf{t}}|\mathbf{s}; \theta) \quad (\text{Self-Powered Generation}) \\ &\doteq -\log P(M(\mathbf{t}, \mathbf{i})|\mathbf{s}; \theta) \quad (\text{Modified Objective}) \end{aligned}$$

where $M(\mathbf{t}, \mathbf{i}) = \arg \max_{\hat{\mathbf{t}}} P(\hat{\mathbf{t}}|\mathbf{t}, \mathbf{i}; \theta)$.

As a result, self-powered model training can improve the model’s responsiveness to instructional inputs and effectively mitigate speech anchor bias.

4 Experiment

4.1 Training Data

We trained the LSM using several well-established ASR datasets: LibriSpeech-960h (Panayotov et al.,

Task	Label Type	#Num.	#Samples
Speech Recognition	Ground-truth	5	237,480
Content Repetition	Self-Powered	5	237,471
Intent Recognition	Self-Powered	5	237,428
Sentiment Analysis	Self-Powered	5	237,434
Keyword Extract	Self-Powered	5	237,353
Continuation	Self-Powered	5	237,391
Speech Translation	Self-Powered	(5,5,5,5)	237,466

Table 1: Statistic of training dataset. “#Num.” refers to the number of instructions per task. We use the En-{De, Zh, Fr, Es} tasks for speech translation.

2015), GigaSpeech-L (Chen et al., 2021), and Common Voice 4.0 (Ardila et al., 2020). Six task types were employed to construct the instruction pool. GPT-4 (OpenAI, 2023) was utilized to generate unique instructional instances for each task, which were then used to generate self-powered data from the ASR datasets. Additionally, we randomly selected 230k samples from the vanilla ASR dataset and integrated them with the self-powered data for model training. Table 1 provides detailed statistics of the training data.

4.2 Training Setup

Self-powered LSM employs the encoder part of Whisper (Radford et al., 2023) model as the speech encoder, and Vicuna-7B-1.5 (Chiang et al., 2023) as the backbone LLM. In the Q-former block, we set $N = 1$ for a single trainable query and $L = 17$ to represent approximately 0.33 seconds per window. We freeze the parameters of the speech encoder and train the Q-former and LLM using a batch size of 512, a learning rate of $2e-5$, a weight decay of 0.05, and warmup steps totaling 100. All the models are trained on eight 80GB A800 GPUs across 2 epoch.

4.3 Evaluation

For ASR, we evaluated the model using the LibriSpeech test-clean and test-other sets, adopting word error rate (WER) as the performance metric. The efficacy of ST was assessed on the CoVoST (Wang et al., 2020) test set for English (En)-{Chinese (Zh), German (De)} translations, and the MuST-C (Di Gangi et al., 2019) tst-COMMON dataset for En-{French (Fr), Spanish (Es)} translations, employing the SacreBLEU score (Post, 2018) as the evaluation. For SLU, we evaluated the model’s capabilities through several tasks. Emotion recognition (ER) was assessed on the MELD dataset (Poria et al., 2019) using the Micro-F1

ID	Method	ASR↓		ST↑		ER↑	KE↑		IC↑	QA↑	
		Clean	Other	CoVoST	MuSTC	MELD	Light	Water	FSC	WebQ	BoolQ
Previous Method											
1	Qwen-Audio* (Chu et al., 2023)	2.6	5.1	24.5	22.0	49.1	0.4	1.1	38.4	71.4	12.2
Implemented Method											
2	Vanilla IT	7.8	13.1	0.2	0.0	0.0	0.0	0.0	0.0	44.3	0.0
3	BLSP (Wang et al., 2023a)	26.6	35.6	8.8	14.6	30.0	4.6	10.3	41.9	71.1	33.6
Our Method with BackBone Model: Vicuna-7B-1.5											
5	Self-Powered LSM with Whisper-small	3.8	8.0	17.5	21.4	51.4	30.7	34.9	47.2	72.4	59.8
6	Self-Powered LSM with Whisper-medium	4.0	7.9	19.2	23.7	49.6	32.3	42.5	58.2	73.4	60.4
7	Self-Powered LSM with Whisper-large	3.3	6.1	20.7	23.8	51.5	36.2	47.1	61.0	73.6	61.4

Table 2: Results span all tasks. “Small”, “Medium”, and “Large” refer to the sizes of the whisper encoders equipped on our method. “*” means we test the open-sources 7B model with the same instruction. The ST outcomes represent an average from En-to-X. More details of ST results are provided in Appendix A.2.

Method	ASR↓		ST↑		ER↑	KE↑		IC↑	QA↑	
	Clean	Other	CoVoST	MuST-C	MELD	Light	Water	FSC	WebQ	BoolQ
Self-Powered LSM	3.8	8.0	17.5	21.4	51.4	30.7	34.9	47.2	72.4	59.8
w/o LLM FT	8.5	14.5	16.8	20.2	35.4	6.9	5.9	76.2	68.5	44.4
w/ LoRA FT	16.1	24.3	11.3	16.0	36.8	25.4	56.7	39.0	70.8	58.8

Table 3: Results span all tasks with different training strategies.

score. Keyword extraction (KE) was evaluated using the SNIP Smart-light and SNIP Washing-machine datasets (Coucke et al., 2019), and intent classification (IC) was tested on the FSC dataset (Lugosch et al., 2019). Accuracy (ACC) served as the metric for both KE and IC tasks. To evaluate the model’s QA capabilities, we utilized TTS³ to convert the Webglm-QA (Liu et al., 2023b) dataset and the BoolQA (Clark et al., 2019a) dataset into a speech format. For the WebQA dataset, BertScore (Zhang et al., 2019) was used as the evaluation metric, and for BoolQA, ACC was used as the evaluation metric. All the instructions used for evaluation are present in Appendix A.3.

4.4 Baseline

We conducted a comparative analysis of the self-powered LSM, benchmarking it against baseline models trained on vanilla ASR speech instructional data as well as previous methodologies, Qwen-Audio (Chu et al., 2023) and BLSP (Wang et al., 2023a). The core idea of BLSP, which involves aligning model behavior using a continuation task, can be seen as an instance of our method. To ensure comparability, we replicated it using identical datasets and architectural frameworks. All the models we evaluated share the same instructions.

³https://www.xfyun.cn/service/offline_tts

4.5 Main Result

Our results, as presented in Table 2, substantiate that our methodology effectively enhances the speech modality capabilities of LLMs for LSMs.

Compared with Vanilla IT Initially, we evaluated a baseline model trained on vanilla speech instructional data (Model “2”), noting substantial overfitting in ASR tasks. This finding supports our hypothesis in Section 3.2 that the direct fine-tuning of LSMs induces significant speech anchor bias, leading to modality imbalance when LLMs are employed to augment the speech capabilities of LSMs. In contrast, training with self-powered data (Model “5”) significantly enhances ASR performance and bolsters the model’s ability to execute instructions across a variety of speech-driven tasks. These findings confirm that our proposed method efficiently imparts speech comprehension and instruction-following abilities to the LSMs.

Compared with Prior Related Studies Compared to Qwen-Audio (Model “1”), which trained using ground-truth data including ASR and CoVoST ST datasets, our method—employing a limited subset of ASR data and no ground-truth translation data—demonstrated superior performance across all SLU and QA tasks. Although this resulted in slightly lower in-domain performance,

Method	ASR↓		ST↑		ER↑	KE↑		IC↑	QA↑	
	Clean	Other	CoVoST	MuST-C	MELD	Light	Water	FSC	WebQ	BoolQ
Emotion + Intent	100.0	100.0	0.0	0.0	71.7	0.0	0.0	99.6	43.2	0.0
Translation	100.0	100.0	4.5	0.0	0.0	0.0	0.0	0.0	42.2	0.0
w/ Self-Powered Aug.	90.8	94.1	10.2	6.2	29.4	15.5	15.1	2.0	50.7	48.6
Self-Powered LSM	3.8	8.0	17.5	21.4	51.4	30.7	34.9	47.2	72.4	59.8

Table 4: Comparison of the LSM training on diverse ground-truth and self-powered datasets.

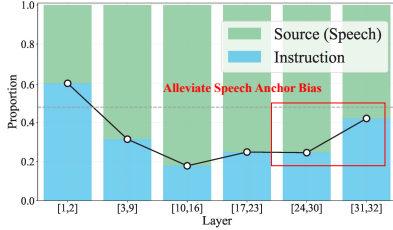


Figure 5: Layer-wise behaviors in self-powered LSM.

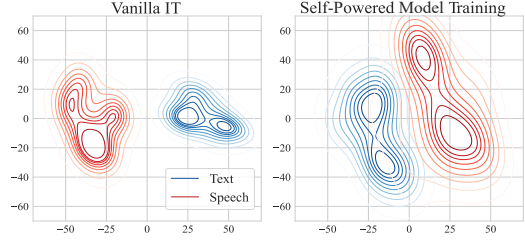


Figure 6: T-SNE visualization of representations from LSMs trained with vanilla IT and self-powered methods.

self-powered LSM excelled in ST tasks on the out-of-domain MuST-C dataset, showcasing its enhanced generalization capabilities derived from training on generated pseudo-data. Compared to models like BLSP (Model “3”), our method exhibited notable improvements in all assessed tasks, particularly in ASR. This underscores our method’s superiority over solely continuation instruction augmentation, highlighting the self-powered augmentation method’s comprehensive enhancement of model training and task performance.

Compared across Different Speech Encoders

To further investigate the impact of varying speech encoder sizes on our method’s efficacy, we employed self-powered LSM across three whisper encoder scales: small, medium, and large-v2. The results (Model “5”, Model “6” and Model “7”) demonstrate that employing more powerful speech encoders enhances LSM performance. With the implementation of whisper-large, self-powered LSM achieves the best results across all evaluated tasks.

5 Analysis

5.1 Ablation Study on Self-Powered LSM

Impact of Different Training Strategy Table 3 details the performance of self-powered LSM under various training strategies. Omitting LLM fine-tuning significantly reduces performance across speech-related tasks, particularly impacting QA capabilities, thus underscoring the crucial role of LLM fine-tuning in achieving instruction alignment. Improvements were observed in SLU and

QA tasks when employing LoRA for LLM. However, performance declined notably in ASR and ST tasks. As discussed by Tang et al. (2023), training the LSM model using fine-tuned LoRA involves intricate design and adjustments of LoRA rank. In contrast, our training method not only simplifies training but also achieves superior performance, demonstrating that full fine-tuning remains the simplest and most superior strategy for expanding LLM with speech capabilities.

Impact of Self-Powered Data To assess the efficacy of self-powered data in augmenting the speech modal capabilities of LLMs, we conducted experiments using various ground-truth and self-powered datasets. As indicated in Table 4, the LSM, when fine-tuned with ground-truth datasets for specific SLU tasks (“Emotion + Intent”) and the CoVoST En-Zh training set (“Translation”), exhibited proficiency only in in-domain tasks. This limitation suggested a lack of instruction-following capabilities, as the model failed to generalize beyond its training conditions. In contrast, training the LSM with self-powered En-Zh translation data of equivalent size endowed the model with nascent instruction generalization abilities. Further extensive training exclusively with fully self-powered data significantly enhanced performance across all tasks.

5.2 Effectiveness of Self-Powered LSM

Mitigating Speech Anchor Bias In Section 3.4, we posit that using self-powered data can enhance

ID	Method	ASR↓		ST↑		ER↑	KE↑		IC↑	QA↑	
		Clean	Other	CoVoST	MuST-C	MELD	Light	Water	FSC	WebQ	BoolQ
	Vanilla IT	40.0	42.6	0.0	0.0	0.2	0.0	0.0	0.0	42.2	0.0
	Self-Powered Llama3	50.9	49.1	15.3	14.1	51.4	56.5	35.8	86.8	67.0	60.6

Table 5: Results of the Llama3-8B-Instruct as the backbone LLM.

Method	ST		SLU		QA	
	Result	Speed	Result	Speed	Result	Speed
Cascade	23.1	1.0×	33.2	1.0×	68.9	1.0×
End-to-End	22.3	3.8×	49.0	3.0×	67.5	3.1×

Table 6: Comparison for performance and speed.

the salience of instructional inputs in LSM training. To substantiate this hypothesis, we conducted a layer-wise behavioral analysis of self-powered LSM. Figure 5 shows that utilizing augmentation data in LSM training markedly improves the model’s sensitivity to instructional cues, especially in the terminal layers (where instruction proportion rises from 0.2 to 0.4). This pattern aligns closely with that observed in instruction-following LLMs, Alpaca and Vicuna, showing that its efficacy in mitigating the speech anchor bias on LSM.

Boosting Speech-Text Alignment We developed a specialized test set of 1,200 examples using five instructional templates (detailed in Appendix A.3) to explore representation alignment between speech and text with the vanilla IT approach and the self-powered LSM. The models processed speech or transcription inputs following identical instructions and extracted averaged representations from the hidden states of the final layer. We applied bivariate kernel density estimation (Parzen, 1962) and utilized the T-SNE technique to reduce data dimensions to a two-dimensional space (Van der Maaten and Hinton, 2008). The findings, depicted in Figure 6, indicate that the self-powered LSM approach significantly improves the proximity between the representations of paired speech and text inputs compared to the vanilla IT method, thereby demonstrating the effectiveness of our methodology in modality fusion.

5.3 Various Model Adapting

To assess the generalizability of our approach, we utilized the Llama3-8B-Instruct as the backbone LLM and implemented self-powered data generated by Llama3-Instruct for training. As shown in Table 5, our method consistently delivers high per-

formance across ST, SLU, and QA tasks. This underscores the effectiveness and adaptability of our self-powered method. Furthermore, we find that both vanilla ASR data and self-powered data contribute minimally to enhancing ASR performance. Although Llama3-Instruct has enhanced capabilities, it often produces additional content and detail, leading to mismatches with the WER evaluation metric in ASR scenarios. We leave this issue for further research to explore deeper evaluations.

5.4 Cascade vs. End-to-End LSMs

We compare our end-to-end LSM with a cascade pipeline. The cascade pipeline employs a two-step approach, initially utilizing Whisper to transcribe speech into text, then leveraging Vicuna-1.5-7B to respond to instructions. Table 6 reveals that although the cascade method can achieve slightly better results in ST and QA, its performance in SLU is limited. In contrast, our end-to-end LSM delivers decent performance on all tasks, with superior inference speed. This distinction highlights the efficiency and potential application benefits of the end-to-end approach. We aim to clarify that the goal of this research is not to outperform the pipeline method but rather to explore the potential of end-to-end models and offer new insights, paving the way for future advancements.

6 Conclusion

In this study, we aim to expand the capabilities of LLMs in speech modalities by exclusively utilizing ASR data. To achieve this, we explore the instruction-following dynamics within LSMs, identifying a significant issue that we term *speech anchor bias*. This bias highlights LSMs’ tendency to unduly emphasize speech inputs. To rectify this, we developed a novel self-powered method that leverages augmented ASR data generated by the models to enhance LSMs’ instruction-following capabilities. Our experiments across multiple speech tasks demonstrate that the self-powered LSM significantly mitigates the speech anchor bias, substantially improving LSMs’ fidelity to instructions.

Limitations

We acknowledge several limitations in our approach to understanding and enhancing large speech models: (1) While we aim for a comprehensive evaluation, our assessments are limited to specific tasks and may lack robustness due to the metrics and datasets used, highlighting the need for further research under broader conditions. (2) Although our method enhances the end-to-end speech capabilities of LSMs, a performance gap remains compared to the cascaded approach, particularly in ST tasks. However, we believe our method provides valuable insights and encourages the development of end-to-end multimodal large models. (3) Concurrent research indicates that carefully curated, smaller, high-quality datasets can significantly enhance a model’s ability to follow instructions (Zhou et al., 2023; Li et al., 2023b; Xu et al., 2023). Despite its size, our dataset offers significant opportunities for refinement. We will make all data publicly available to promote ongoing academic exploration.

Ethics Statement

In conducting our research, we strictly adhere to all applicable laws and regulations. Additionally, we take ethical considerations seriously and adhere to the standards of the ACL Ethics Policy. This paper focuses on understanding and improving large speech model. To support reproducibility, both the datasets and code used in this study are available to other researchers upon request. We have made every effort to present our findings and conclusions in an objective and accurate manner.

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Model	CoVoST		MuST-C	
	en-zh	en-de	en-es	en-fr
Qwen-Audio	30.8	18.2	20.7	23.2
Vanilla IT	0.1	0.2	0.0	0.0
BLSP	10.8	6.7	12.2	17.0
Self-Powered LSM Small	20.2	14.8	18.9	23.9
Self-Powered LSM Medium	22.7	16.2	21.1	26.3
Self-Powered LSM Large	23.9	17.4	20.7	26.8

Table 7: CoVoST-2 and MuST-C ST results.

A Appendix

A.1 Attention as Explanation

Formulation Attention weights are a widely used interpretation tool that identifies patterns in attention interactions to determine which input vectors are most influential. Each self-attention weight $\alpha_{i,j}$ is computed from the corresponding input $\mathbf{X} = [\mathbf{x}_i]_{i=1}^I \in \mathbb{R}^{I \times d}$:

$$\alpha_{i,j} = \underset{\mathbf{x}_j \in \mathbf{X}}{\operatorname{softmax}} \left(\frac{q(\mathbf{x}_i) k(\mathbf{x}_j)^\top}{\sqrt{d}} \right) \in \mathbb{R}, \quad (5)$$

where $q(\cdot)$, $k(\cdot)$ are the query and key transformations, respectively. For multi-modal inputs, the features themselves also affect critical token interactions. We define the *norm of the weighted transformed vector* as $\mathbf{a}(i, j)$, which compute from the self-attention weight $\alpha_{i,j}$ and the j -th input vector $\mathbf{x}_j$:

$$\mathbf{a}(i, j) = \|\alpha_{i,j} (v(\mathbf{x}_j) \mathbf{W}^O)\|, \quad (6)$$

where $v(\cdot)$ is value transformations, $\mathbf{W}^O$ was introduced in Vaswani et al. (2017) which integrate the output of multi-heads applied to each attention. Here, $\mathbf{a}(i, j)$ can represent the norm-based attention score from the j -th word to the i -th word. When the model generates a M -sized text for every layer, we compute the average attention scores $\mathbf{A}$:

$$\mathbf{A}_j = \frac{1}{M} \sum_{m=1}^M \mathbf{a}(m, j). \quad (7)$$

Here, $\mathbf{a}(m, j)$ represents the attention score from the j -th token to the m -th generated token, with $\mathbf{A}_j$ denoting the average attention score from the j -th token to the model’s output.

Experiments Setup We conduct a comparative analysis of layer-wise behavior between LLMs and LSMs regarding their handling of instruction and source inputs. For LLMs, our analysis includes Llama2-base (Touvron et al., 2023b), Vicuna-1.5-7B (Chiang et al., 2023), and Alpaca-7B (Taori

et al., 2023). For LSMs, we evaluate an untrained model, a model trained on the ASR task using the LibriSpeech-960h dataset, and a model trained on En-Zh ST task using the CoVoST2 dataset. Each model comprises 32 layers, which we categorize and average into six levels for enhanced visualization clarity.

A.2 Details of Experimental Results

Detail ST Results Table 7 showcases the outcomes of our ST results for a range of language pairs, including CoVoST En-{De, Zh}, as well as MUST-C English-to-{Es, Fr} ST tasks. The result shows that IAFT consistently exceeds the performance benchmarks set by existing baseline models across all the specified ST tasks, confirming the efficacy of our method.

A.3 Details of Our Instructions

Instructions Used for Visualization The instructions used for attention and speech-text representation visualization are presented in Table 8. To enhance the reliability of our analysis, we diversify it by including five distinct instruction types, covering a broad spectrum of tasks: speech translation, automatic speech recognition, speech summarization, keyword extraction, and question generation. For each of these categories, we randomly select 240 samples from three different datasets: LibriSpeech test set, the CoVoST2 En-Zh test set, and the MuST-C En-Zh tst-COMMON dataset, resulting in a total test set of 1,200 samples.

Instructions Used for Evaluation The instructions used for evaluation are presented in Table 9. For all LSMs evaluated, we use the same instructions to ensure a fair comparison.

Instructions Used for Data Generation The instructions utilized for data generation are detailed in Table 10. We employ six distinct types of tasks spanning various tasks: content repetition, keyword extraction, intent recognition, sentiment analysis, continuation, and translation. These instructions are used to prompt LLM to generate self-powered data and to expand the speech ability of LLM. For the ground-truth ASR dataset, the instructions we used are presented in Table 11.

Speech Translation

Provide the translation text from English to Chinese according to the speech.

Speech Recognition

Provide the transcription according to the speech.

Speech Summarization

Listen to the speech and provide a summary, capturing the main points in no more than three sentences.

Keyword Extraction

Identify and list the keywords or phrases from the speech, focusing on the most relevant terms used.

Question Generation

Based on the content of the speech, generate a relevant question.

Table 8: Instructions used for attention and representation analysis

Speech Recognition

Provide the transcription according to the speech.

Speech Translation

Provide the translation text from {source} to {target} according to the speech. (Do not generate extra information)

Emotion Recognition

Classify the emotion of the speech from {'neutral', 'joy', 'sadness', 'anger', 'surprise', 'fear', 'disgust'}. Ensure your response strictly adheres to this format: {'xxx'}.

Intent Recognition

Classify one of the intent label in ['activate lamp', 'activate lights', 'activate music', 'bring juice', 'bring newspaper', 'bring shoes', 'bring socks', 'change language Chinese', 'change language English', 'change language German', 'change language Korean', 'change language none', 'deactivate lamp', 'deactivate lights', 'deactivate music', 'decrease heat', 'decrease volume', 'increase heat', 'increase volume'] according to the speech.

Keyword Extraction - Light

Please listen carefully to the SPEECH provided and extract two keywords from the following list: 'bedroom', 'brightness', 'decrease', 'increase', 'kitchen', 'living room', 'turn off', 'turn on'. Your response should strictly follow this format: ['keyword1', 'keyword2'].

Keyword Extraction - Water

Please listen carefully to the SPEECH provided and extract three keywords from the following list: 'bedroom', 'brightness', 'decrease', 'increase', 'kitchen', 'living room', 'turn off', 'turn on'. Your response should strictly follow this format: ['keyword1', 'keyword2', 'keyword3'].

BoolQA

Answer the questions in speech based on the CONTEXT given, your answer is only true or false, you don't need to answer anything else. CONTEXT: {references}

WebQA

Answer the questions in speech based on the CONTEXT given: CONTEXT: {references}

Table 9: Instructions used for evaluation.

Content Repetition

1. Repeat the provided text, ensuring to maintain its original meaning and details
2. Rephrase the text without altering its initial intent and key information
3. Paraphrase the provided text while preserving all original facts and nuances
4. Echo the content of the text, maintaining its exact purpose and details
5. Retell the given information without changing its meaning or losing any critical data

Translation

1. Provide the translation from English to {target}
2. Translate the given English content into {target}
3. Render the English text into {target}
4. Convert the specified English text into {target}
5. Translate the provided English material into the {target} language

Keyword Extraction

1. Extract the most frequently occurring words or phrases in the text, excluding common stopwords, to identify main topics
2. Identify and list the most common words or phrases from the text, omitting typical stopwords, to highlight central themes
3. Determine the key words or phrases frequently used in the text, removing all usual stopwords, to discern the main topics
4. Find the recurring words or phrases in the text, ignoring common stopwords, to ascertain the primary themes
5. Extract significant words or phrases that appear often in the text, exclude basic stopwords, to uncover the main subjects

Intent Recognition

1. Determine the primary purpose of the speech and evaluate how clearly and effectively the message is conveyed
2. Identify the main intent of the speech and assess the clarity and effectiveness of its delivery
3. Ascertain the fundamental objective of the speech and critique the transparency and efficiency of its presentation
4. Figure out the aim of the speech and judge how lucidly and effectively the ideas are presented
5. Assess the central purpose of the speech and evaluate the directness and impact of its expression

Sentiment Analysis

1. Determine the sentiment of the text and identify which sections contribute most to sentiment
2. Analyze the overall mood of the text and pinpoint the parts that heavily influence the sentiment
3. Evaluate the emotional tone of the text and determine which segments primarily affect the sentiment
4. Identify the feeling conveyed by the text and specify which portions substantially shape this sentiment
5. Assess the sentiment expressed in the text and highlight which areas contribute most to this feeling

Continuation

1. Please write a coherent and engaging English continuation of the given English text with less than 50 words
2. Compose a logical and captivating follow-up to the provided English text within 50 words
3. Craft a coherent extension for the English text, ensuring it does not exceed 50 words
4. Develop a consistent and attractive continuation of the English text, keeping it under 50 words
5. Write a fluent and engaging continuation of the English text, limited to 50 words

Table 10: Instructions used for data generation.

Speech Recognition

1. Provide the English transcription according to the speech
2. Transcribe the spoken words into written English text
3. Convert the spoken English into a written transcript
4. Create a written transcription of the spoken English
5. Write down the English speech as a text transcript

Table 11: Instructions used for ground-truth ASR datasets.